

Quantum Field Theory – Homework 6

Name: _____ Due Date: 2025.12.01

The following problems are based on Lectures 11 and 12, focusing on propagators, contour prescriptions, scattering theory, and the derivation of path integrals from time-slicing. Follow the structure of the hints and show all essential steps.

Problem 1. Momentum Conservation and the Structure of the S-Matrix (20 points)

In Lecture 11, the S -matrix was defined as the transition amplitude between asymptotic free states:

$$S_{\beta\alpha} = \langle \beta, +\infty | \alpha, -\infty \rangle.$$

- (a) Explain why translational invariance of the Hamiltonian implies conservation of the total 4-momentum. Hint: use $[P^\mu, H] = 0$ and the fact that P^μ generates translations.
- (b) Show that the interaction part of the S -matrix must take the form

$$S_{\beta\alpha} = \delta_{\beta\alpha} + i(2\pi)^4 \delta^{(4)}(p_\beta - p_\alpha) M_{\alpha \rightarrow \beta}.$$

Identify clearly which part is “kinematic and which part contains the actual physics of scattering.

- (c) For a $2 \rightarrow 2$ scattering process, write explicitly the momentum-conservation delta-function in terms of the four external momenta.

Problem 2. Deriving the Path Integral for the Harmonic Oscillator (10 points)

Consider the non-relativistic harmonic oscillator with Hamiltonian

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 x^2.$$

By following the time-slicing steps in Lecture 12, show the path-integral representation of the propagator explicitly

$$K(x, t; x', t') = \langle x, t | x', t' \rangle = \int_{x(t')=x'}^{x(t)=x} \mathcal{D}x(t) \exp \left[i \int_{t'}^t d\tau \left(\frac{1}{2}m\dot{x}^2(\tau) - \frac{1}{2}m\omega^2 x^2(\tau) \right) \right].$$